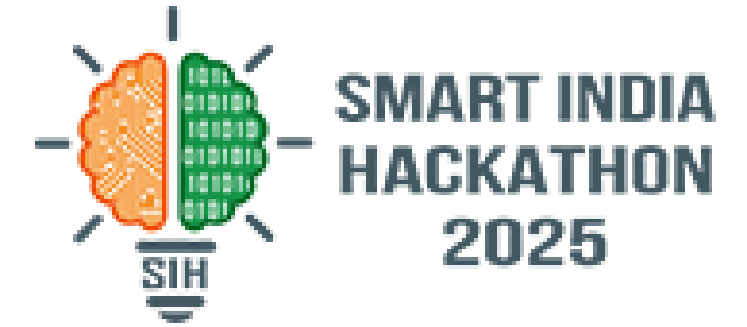


SMART INDIA HACKATHON 2025

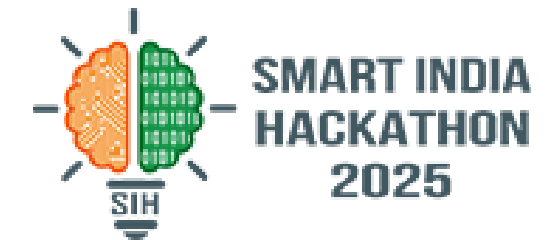


- **ID :** 25131
- **Title :** Student Innovation
- **Theme :** MedTech / BioTech / HealthTech
- **Category :** Software
- **Team ID :**
- **Team Name :** Rakshanu





PROBLEM STATEMENT



Cutting-edge technology in these sectors continues to be in demand. Recent shifts in healthcare trends, growing populations also present an array of opportunities for innovation.

Problems

Unpredictable Viral Mutations

Current systems are slow to track and predict mutations.

Delayed Drug Discovery

Traditional lab-based drug testing takes months/years.

Reactive Healthcare

Outbreaks detected only after spread, no early-warning system.

Lack of Context

Viral prediction ignores real-world vitals/environmental factors.

Solution

Viral Insight & Rapid Optimization Analytics Intelligence

Mutation Prediction

AI analyzes genome and protein data to identify and highlight potential viral mutations.

Drug Candidate Suggestion

A cheminformatics engine evaluates and ranks promising antiviral compounds for treatment.

Dashboard Visualization

Interactive dashboards display mutation hotspots, candidate drugs, 3D protein structures, and associated risk indices.

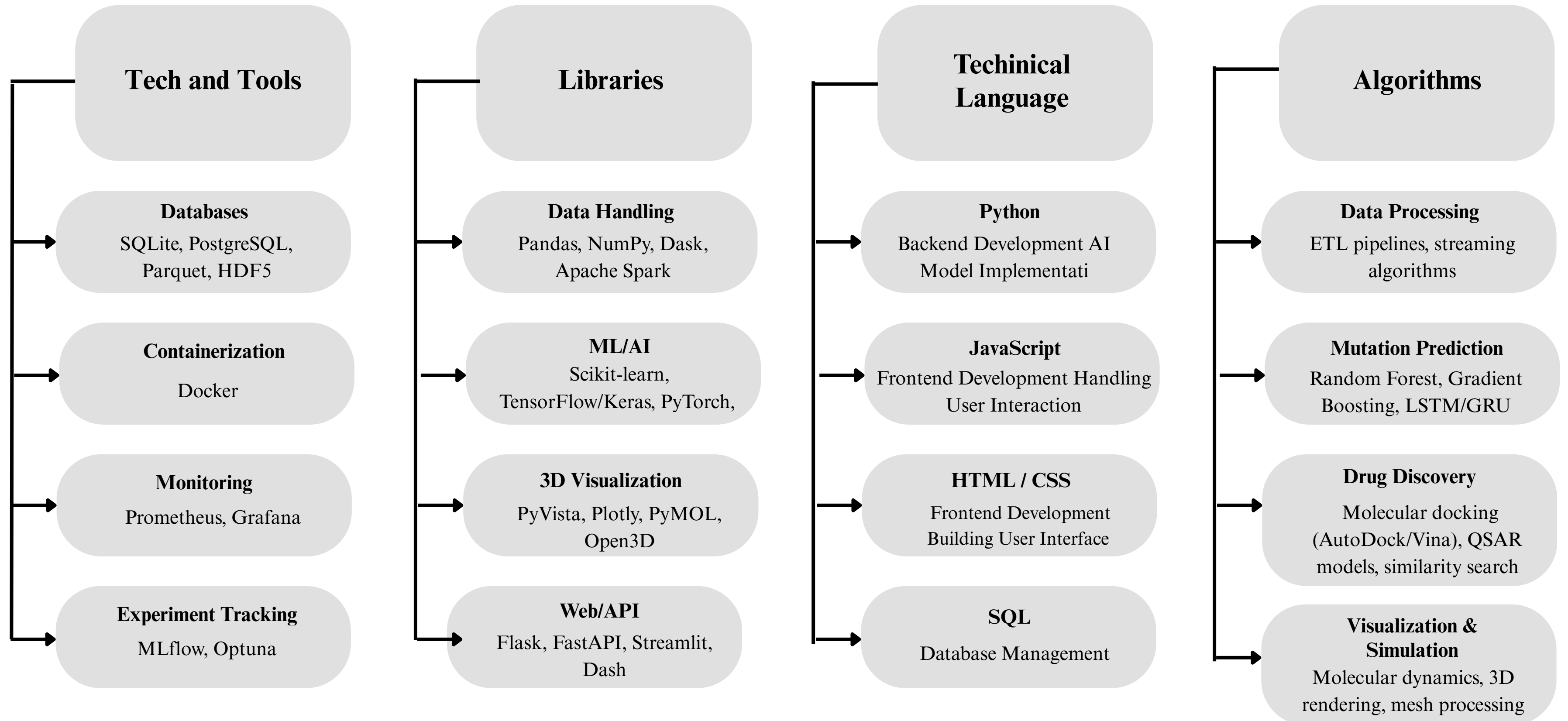
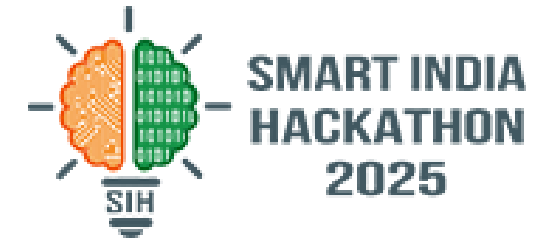
Proactive Threat Management

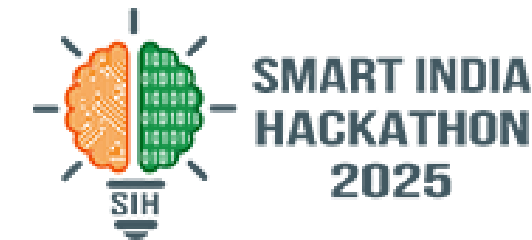
Insights from **data-driven** predictions help healthcare systems prepare and respond to emerging viral threats effectively.

Viro-AI is a powerful, **AI-driven** platform that predicts viral mutations and suggests potential drug candidates to accelerate outbreak response and research.

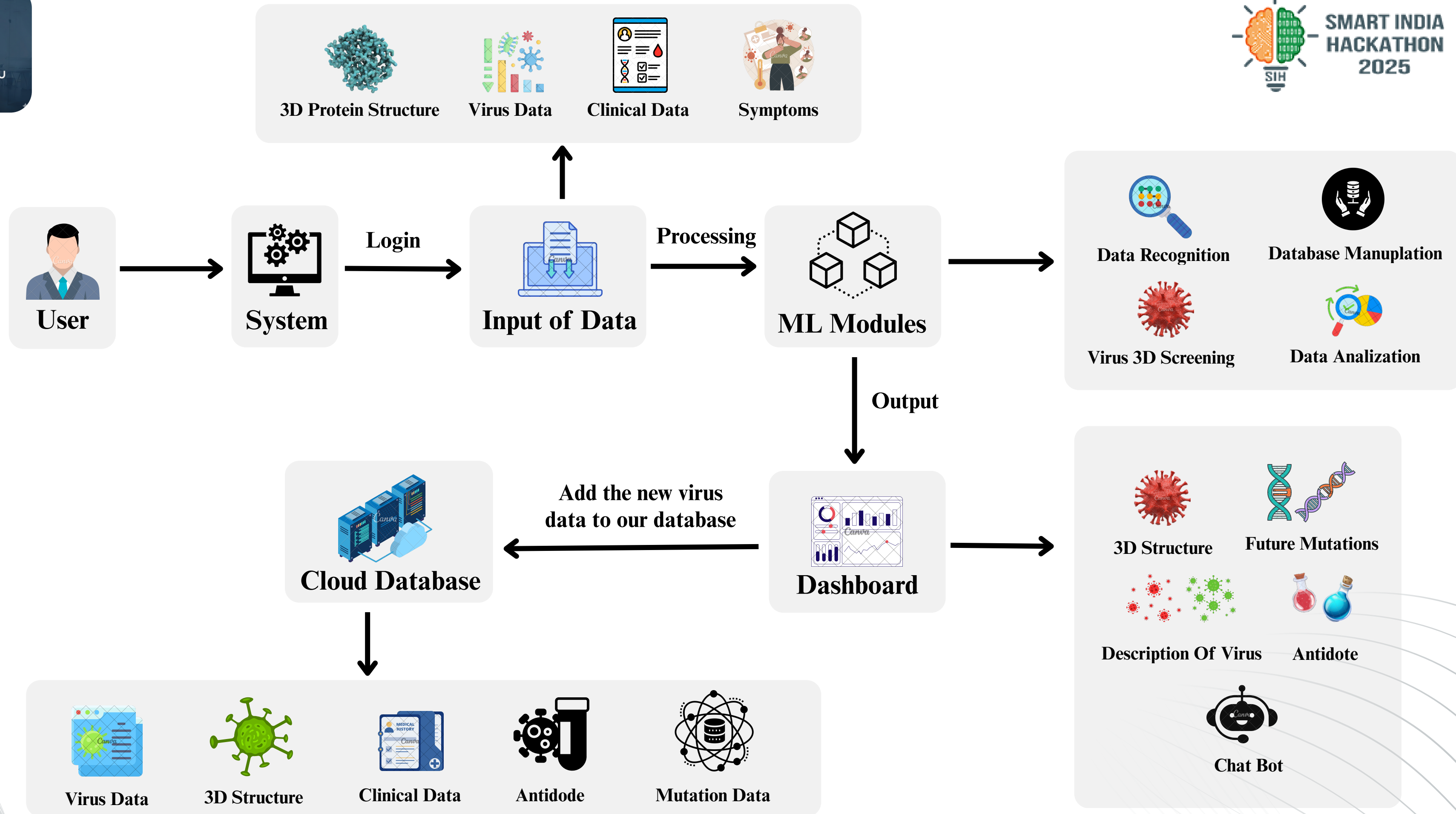


TECHNICAL APPROACH



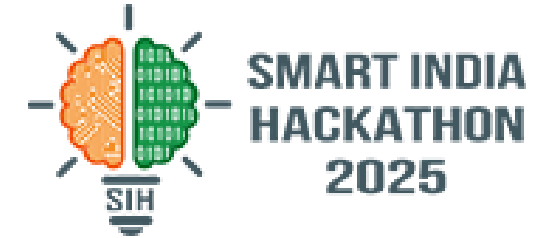


SYSTEM ARCHITECTURE





FEASIBILITY & VIABILITY



Strengths & Readiness

Architecture

Built on a scalable, cloud-based architecture with open-source technologies.

Data Strategy

Leverages a continuous stream of public data for model training and validation

Production Readiness

Emphasizes a maintainable structure for large-scale, real-world deployment.

Potential Challenges

Regulatory Compliance

Navigating global health data regulations is a critical challenge.

Trust & Transparency

Risk of algorithmic bias could erode trust; requires model explainability.

Operational Costs

Managing high data volume and requests efficiently while being cost-effective.

Strategies

Security

Implement a proactive security framework with encryption and access controls.

Explainability

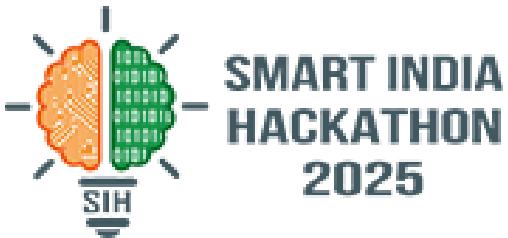
Integrate Explainable AI (XAI) to provide transparent reasoning for decisions.

Scalability

Use dynamic cloud infrastructure with containerization to optimize performance and costs.



IMPACT & BENEFITS



Benefits

Social

Early detection and faster response save lives during outbreaks.

Economic

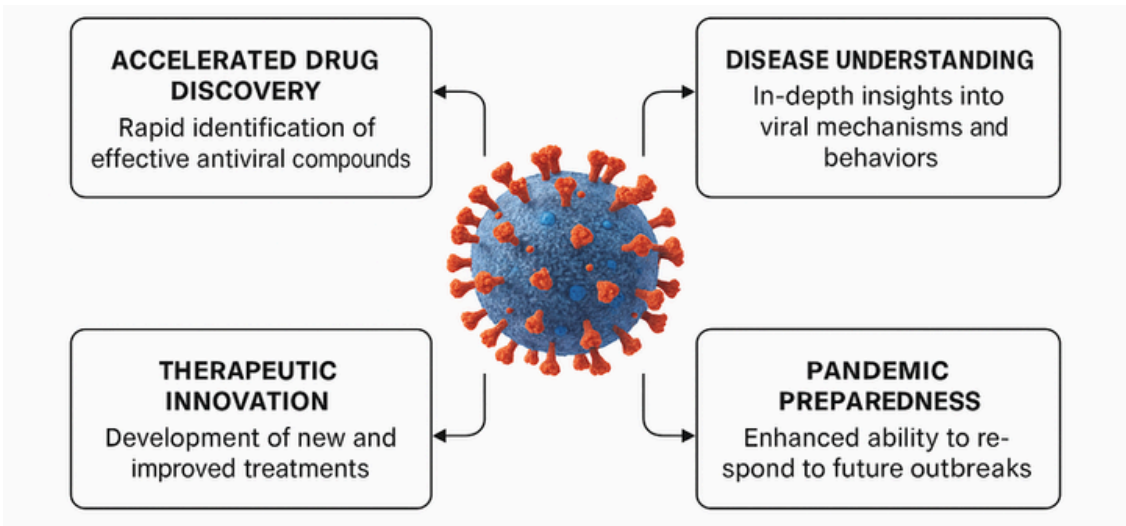
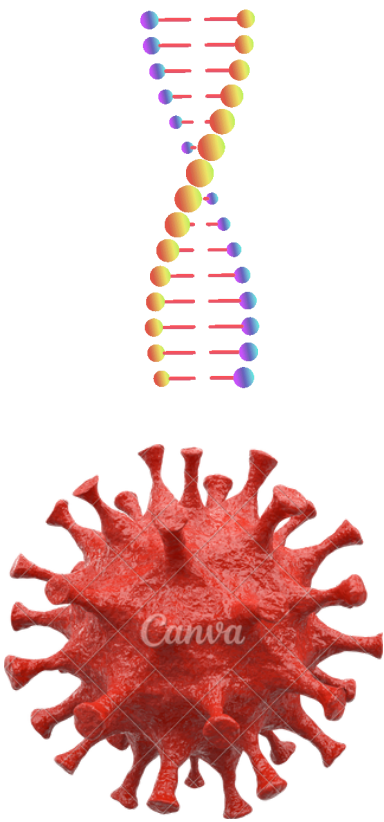
Cuts down cost and time in drug discovery using AI + Cheminformatics.

Healthcare

Provides clear visual insights to support medical professionals.

Technological

Integrates AI, ML, Cheminformatics, and 3D Visualization into one solution.



Impact

Accelerated Research

AI/ML models speed up genome and protein analysis, reducing months of work to hours.

Proactive Response

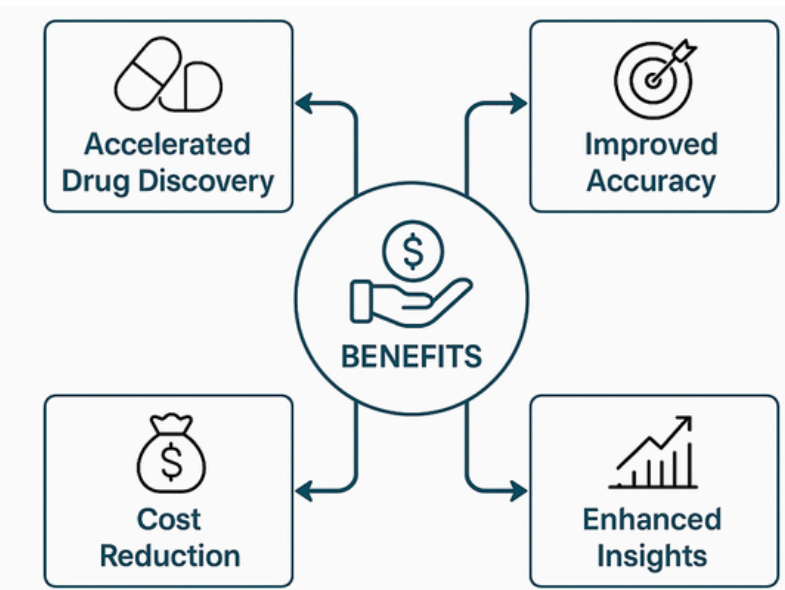
Predicting mutation hotspots enables faster vaccines and treatments.

Healthcare Support

Dashboard simplifies bioinformatics, helping labs and hospitals take timely decisions.

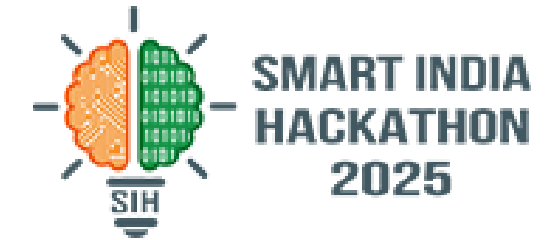
Global Preparedness

Scalable system with potential for LLM-based biomedical reasoning to improve pandemic readiness.





RESEARCH & REFERENCES



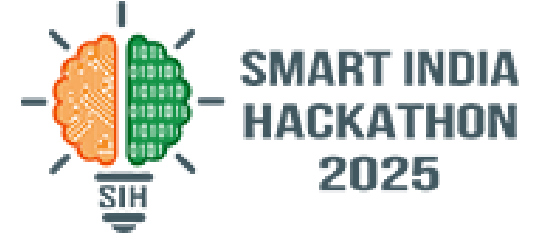
Sr. No	Author	Paper Title	Publication	Year	Link
1)	Jumper, J. et al.	AlphaFold: Highly accurate protein structure prediction	Nature	2021	https://www.nature.com/articles/s41586-021-03819-2
2)	Kokudeva, M. et al.	AI as a tool in drug discovery and development	Biotechnology & Biotechnological Equipment	2021	https://www.nature.com/articles/s41573-019-0016-z
3)	Deng, J. et al.	Artificial intelligence in drug discovery: applications & techniques	Briefings in Bioinformatics	2020	https://www.nature.com/articles/s41573-019-0016-z
4)	Vamathevan, J. et al.	Machine learning in drug discovery and development	Nature Reviews Drug Discovery	2019	https://www.nature.com/articles/s41573-019-0016-z



“For more details about this project, please scan the QR code to access a detailed overview.”



TEAM MEMBER DETAILS



College Name : Dr. D. Y. Patil College of Engineering & Innovation, Varale, Talegaon, Pune - 410507

Team Leader Name : Sairaj Popat Jadhav

Branch : Bachelor of Engineering

Stream : Artificial Intelligence & Data Science

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Team Member 2 Name : Pranav Nitin Shinde

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Stream : Computer Engineering

Year : TE

Team Member 4 Name : Yash Yogesh Wase

Branch : Bachelor of Engineering

Stream : Artificial Intelligence & Data Science

Year : TE

Team Member 5 Name : Janvi Amol Patil

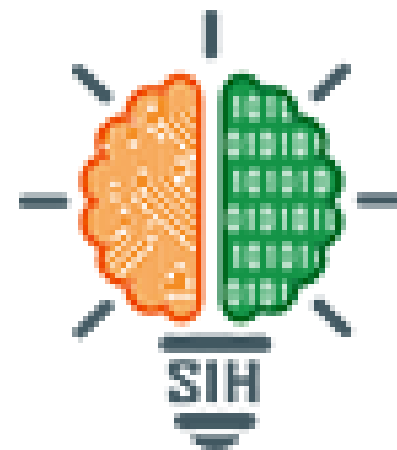
Branch : Bachelor of Engineering

Stream : Artificial Intelligence & Data Science

Year : TE



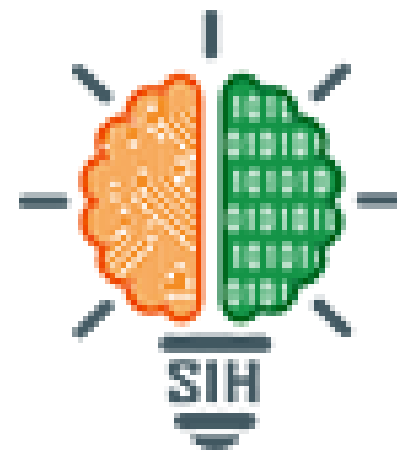
ANY QUESTIONS ??



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THANK YOU



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